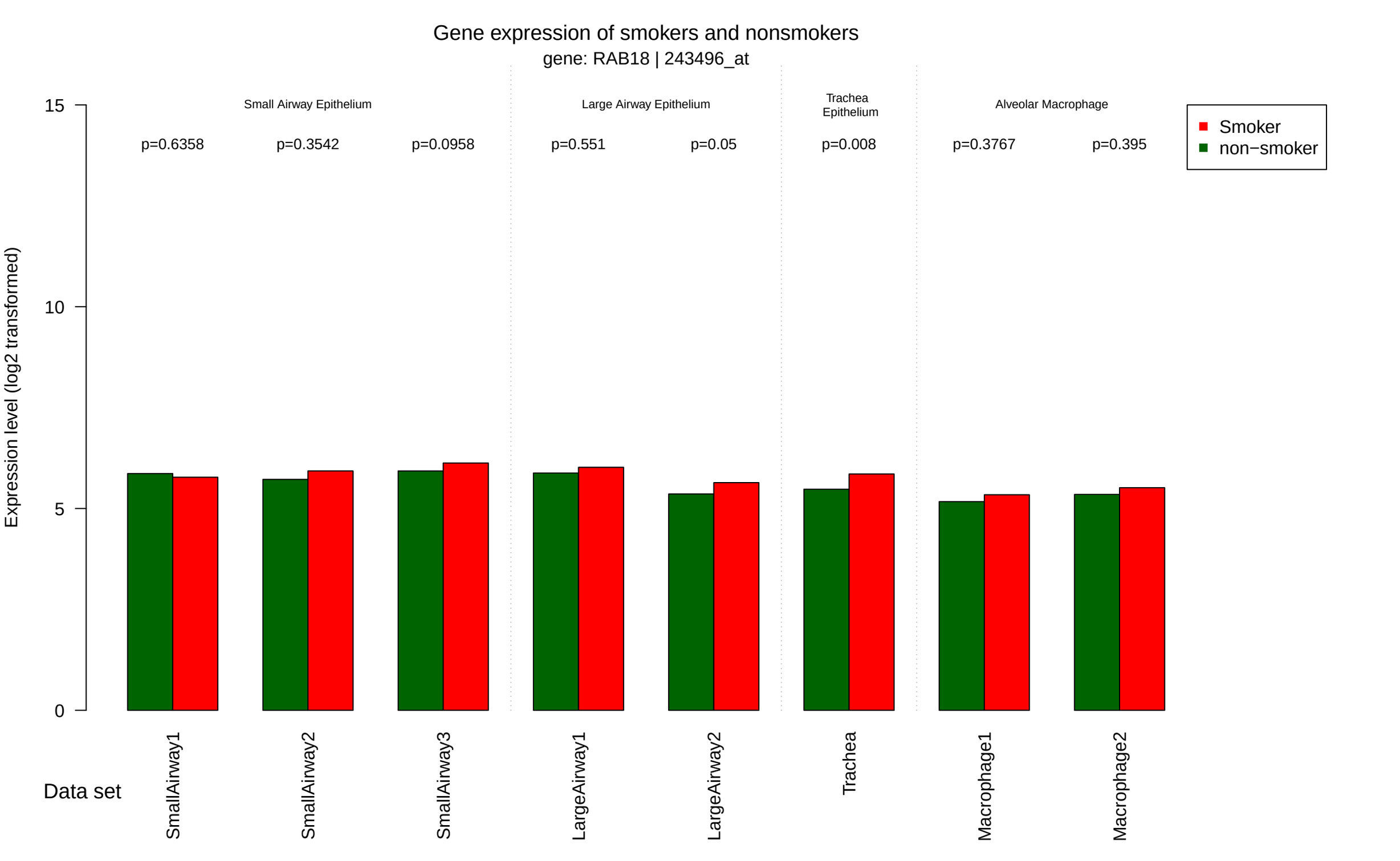
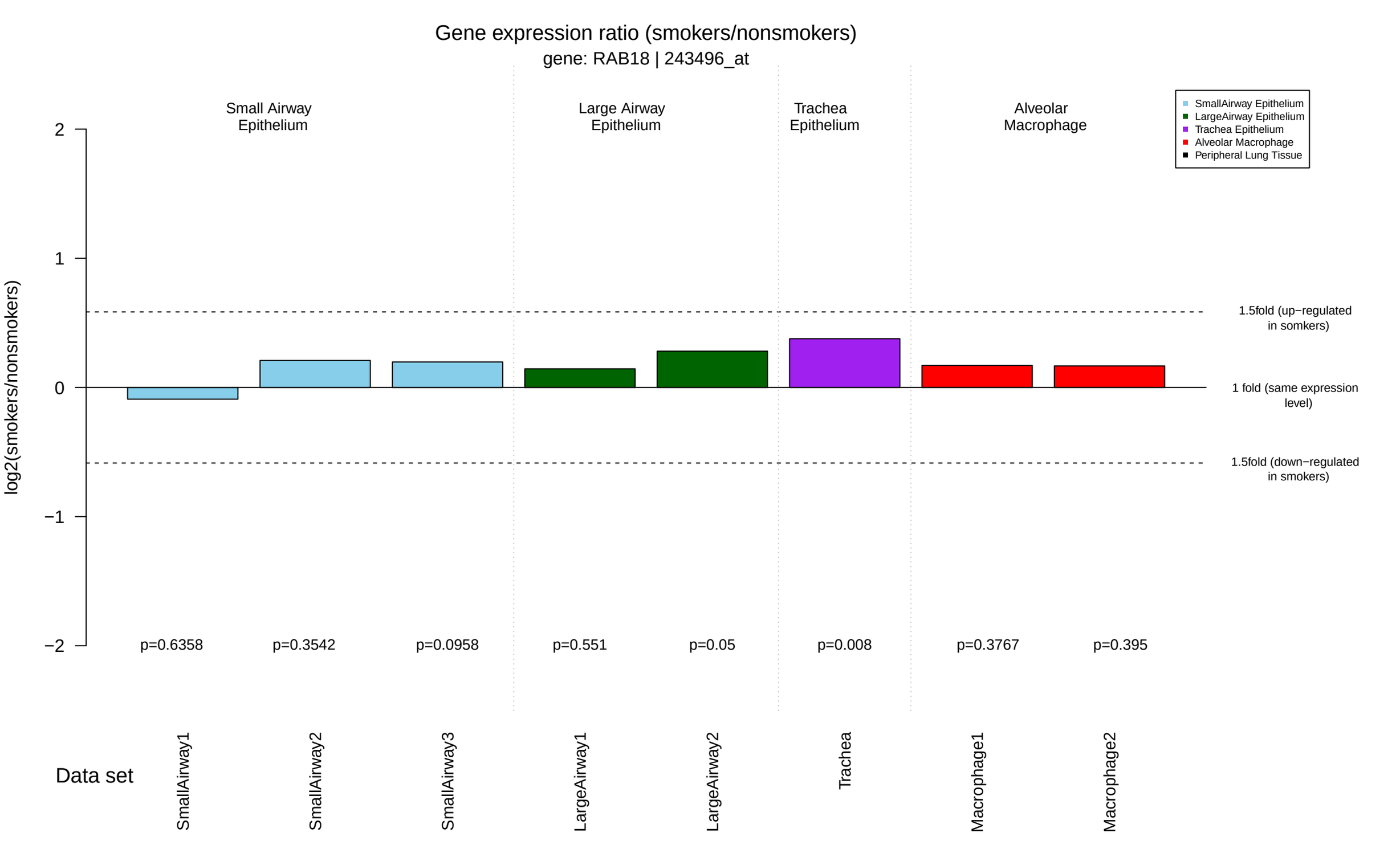
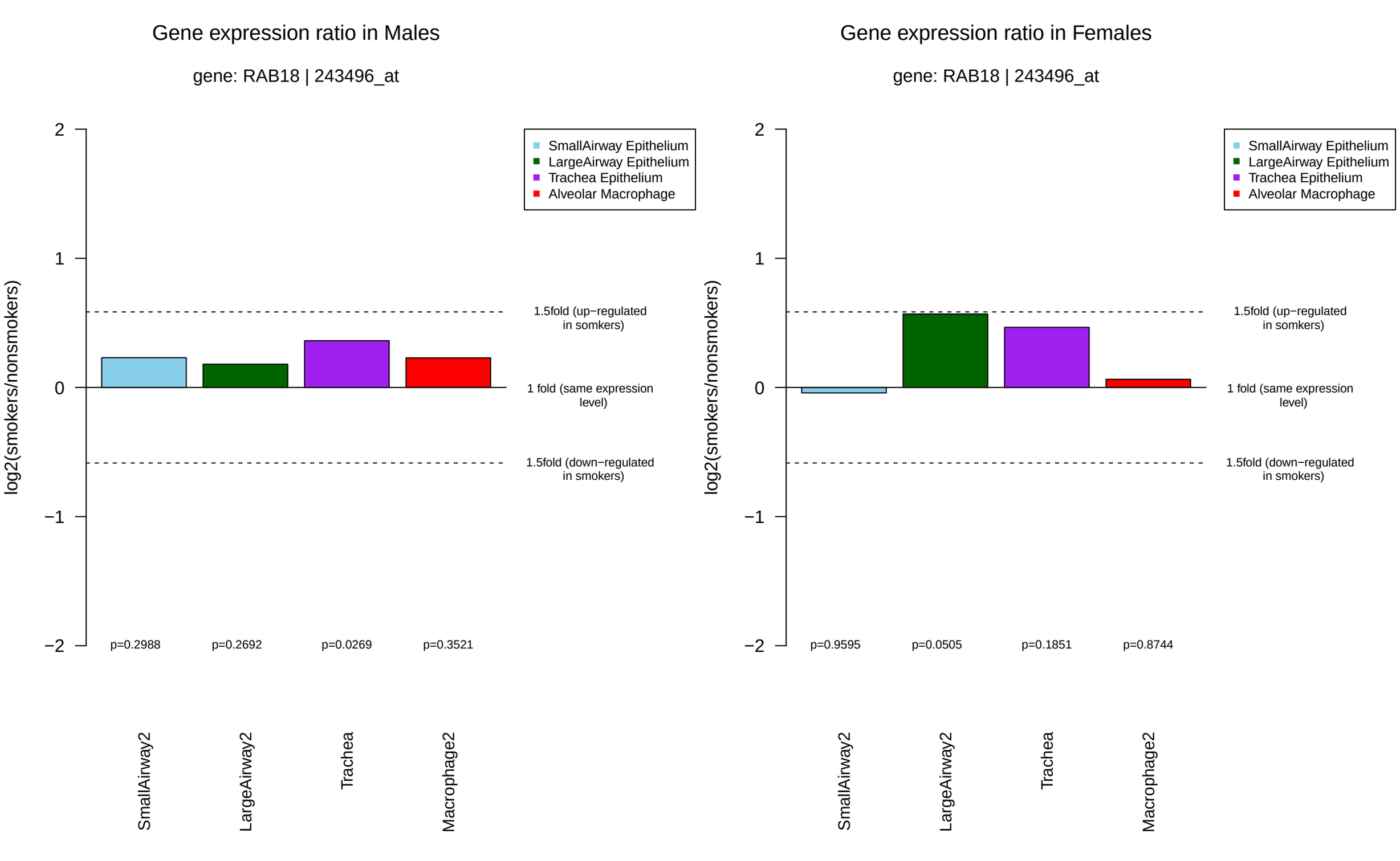
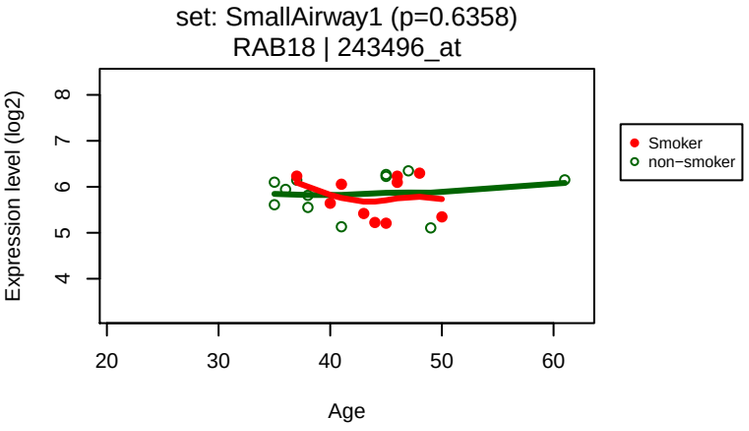
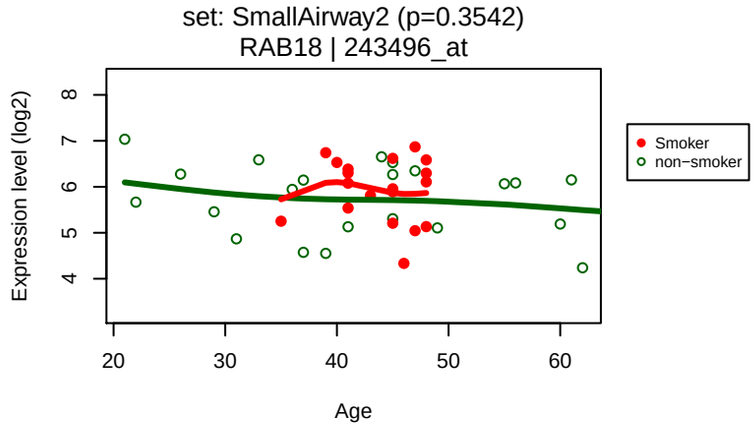
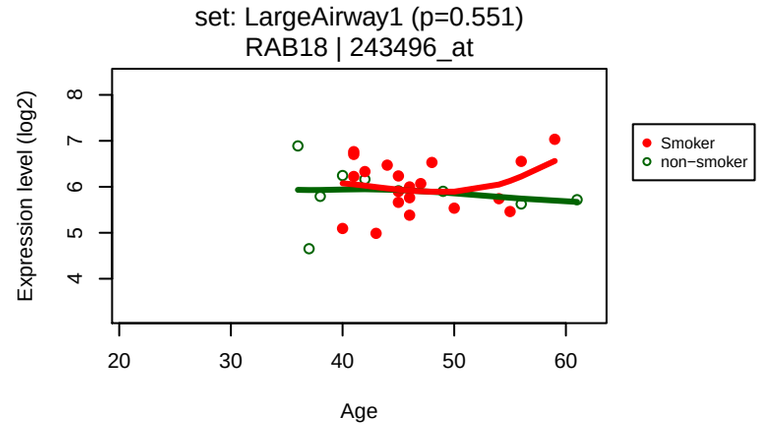

Gene expression of smokers and nonsmokers
gene: RAB18 | 243496_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6358
p=0.3542
p=0.0958
p=0.551
p=0.05
p=0.008
p=0.3767
p=0.395
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

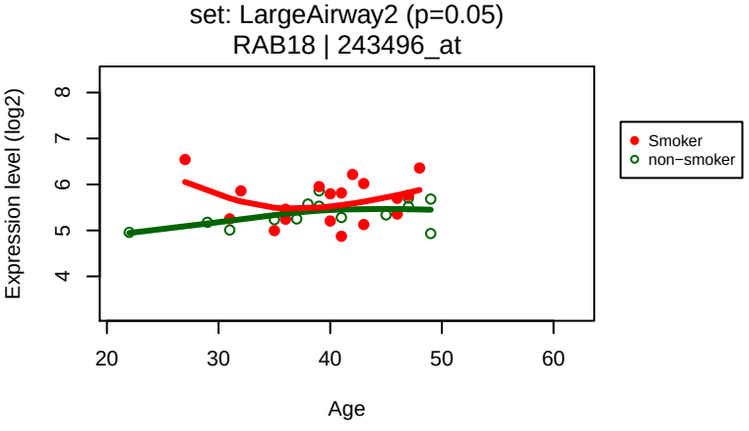

Gene expression ratio (smokers/nonsmokers)
gene: RAB18 | 243496_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6358
p=0.3542
p=0.0958
p=0.551
p=0.05
p=0.008
p=0.3767
p=0.395
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

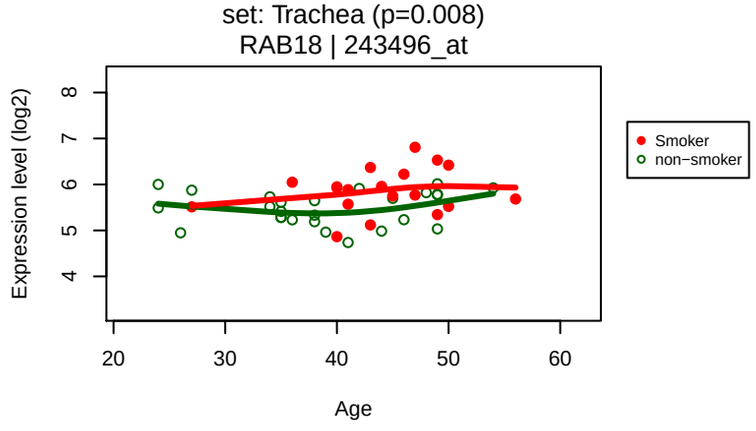

Gene expression ratio in Males
gene: RAB18 | 243496_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2988
p=0.2692
p=0.0269
p=0.3521
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB18 | 243496_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9595
p=0.0505
p=0.1851
p=0.8744
SmallAirway2
LargeAirway2
Trachea
Macrophage2

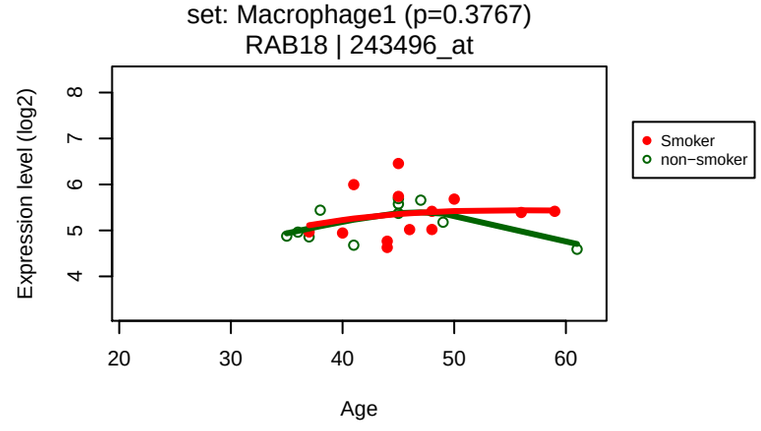

set: SmallAirway1 (p=0.6358)
RAB18 | 243496_at
Expression level (log2)
Age
Smoker
non−smoker

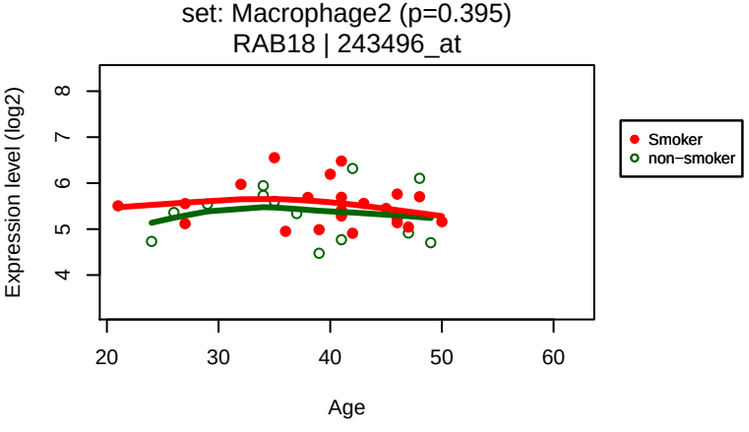

set: SmallAirway2 (p=0.3542)
RAB18 | 243496_at
Expression level (log2)
Age
Smoker
non−smoker

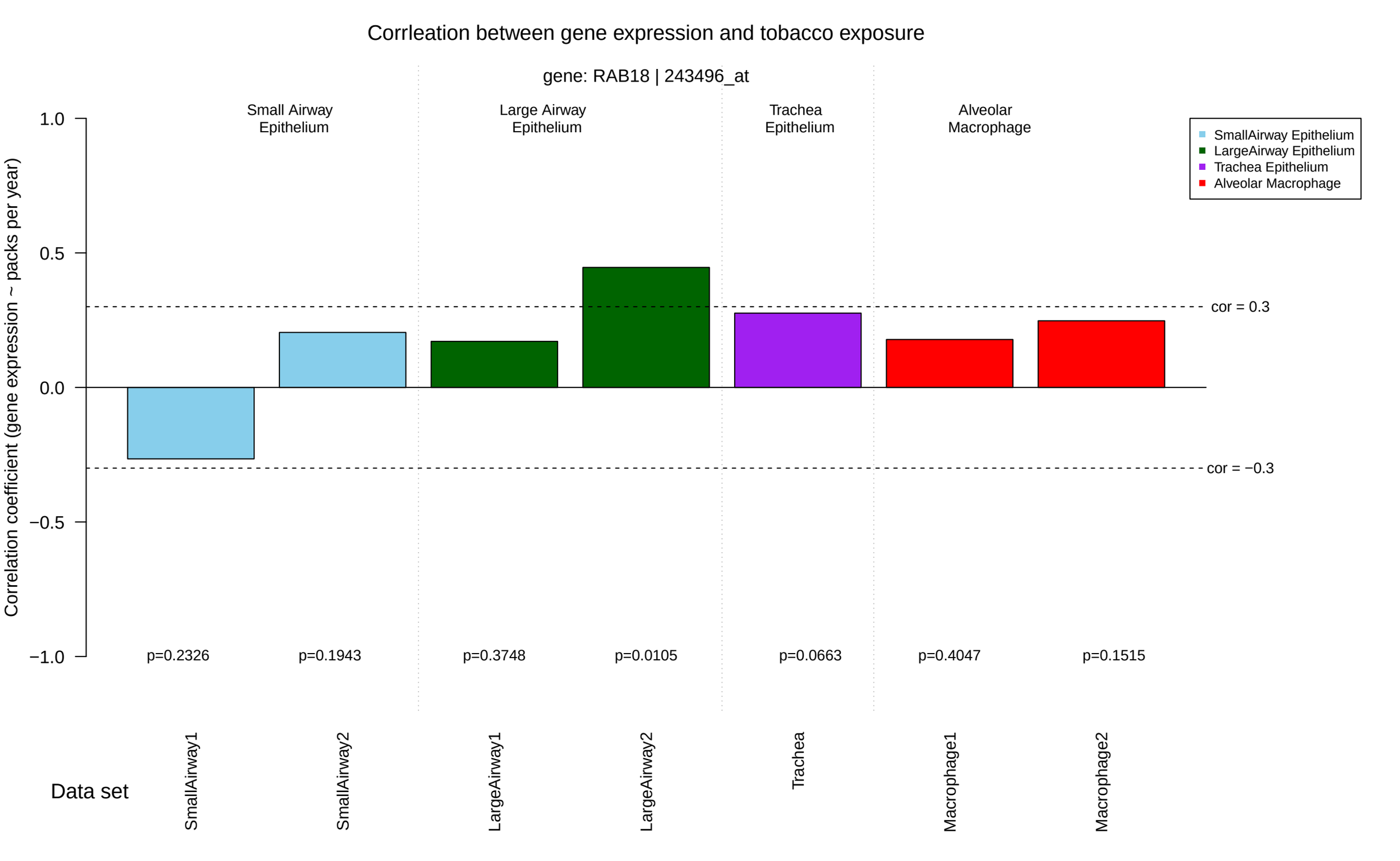

set: LargeAirway1 (p=0.551)
RAB18 | 243496_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.05)
RAB18 | 243496_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.008)
RAB18 | 243496_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3767)
RAB18 | 243496_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.395)
RAB18 | 243496_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB18 | 243496_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2326
p=0.1943
p=0.3748
p=0.0105
p=0.0663
p=0.4047
p=0.1515
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2